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Meshing procedure for modelling of atmospheric wind flow

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summary

1. Pre-processing inputs
2. Targets
3. Mesh construction
4. OpenFOAM tools

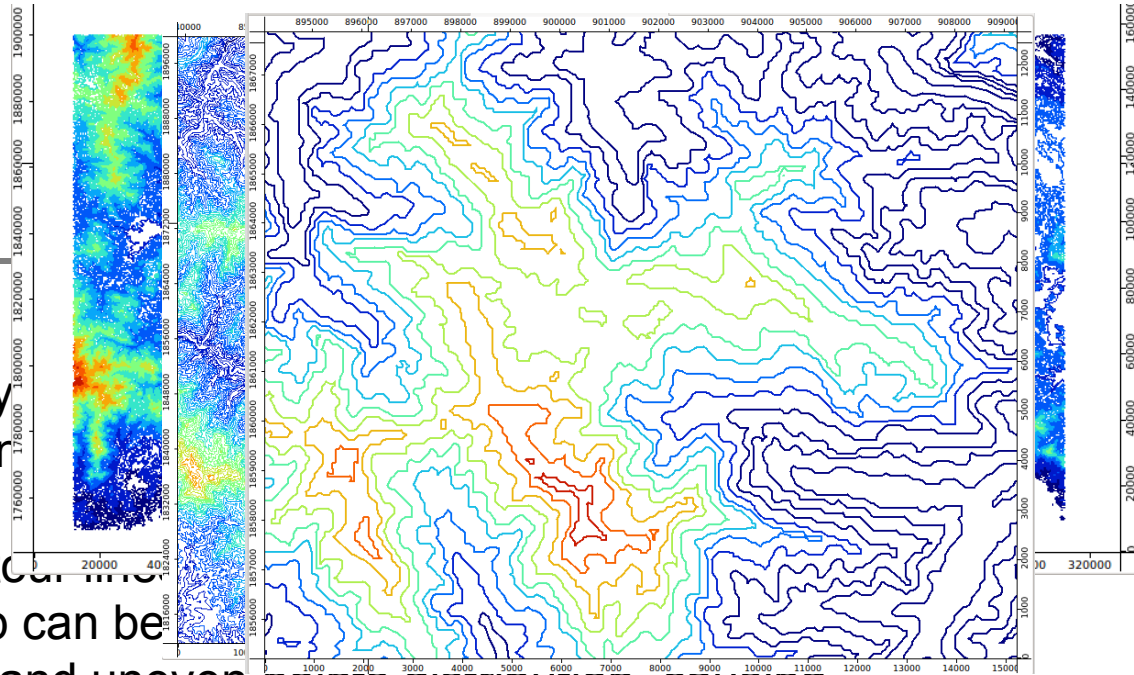
The background of the slide is composed of several overlapping, semi-transparent geometric shapes. On the left, there is a large, dark blue triangular shape pointing towards the center. To its right, a lighter blue shape overlaps it. Further right, there are green and yellow shapes. At the bottom, there are magenta and red shapes. A large, light gray shape is positioned in the center-right, partially overlapping the other colors. The overall effect is a modern, abstract design.

1. Pre-processing
inputs

.map files

1. Pre-processing inputs

- .map files is an industry and for model validation
- Topography
 - Encoded with contour lines
 - Resolution of .map can be
 - Large white spots and uneven points distribution, causing problems with terrain approximation.
 - Near-shore topographies often come without the shore line being marked.



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.map files

1. Pre-processing inputs

- .map files is an industry standard. It is used (and enforced) by clients and for model validation benchmarks.
- Topography
- Roughness
 - Encoded with regions (curve with inside and outside value of roughness).
 - Always broken and need manual editing.
 - Maps are often cropped (chopped) and sometimes nested.

.map files almost always need manual editing (WAsP editor + satellite pictures from Google Earth).



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1. Pre-processing inputs

- Delaunay triangulation
 - Very complex to implement
 - Always works, with good results
 - Very slow
- Shepard (distance weighted average of nearest neighbors)
 - Simple to implement and easy to speed up
 - Can perform smoothing
 - Have problems with bad .map files



Terrain interpolation

1. Pre-processing inputs

- Complexity
 - For $n = \text{no_input_data}$, and $m = \text{points_to interpolate}$
 - Delaunay: triangulation - $O(n \log n)$, sampling – $O(n*m)$
 - Shepard: sampling $O(m*n)$
 - Improved Shepard: initialization - $O(n)$, sampling – $O(m*\log n)$
 - For a typical case where $n=m=100\ 000$
 - Delaunay: $O(10^{10})$
 - Shepard: $O(10^6)$ – faster about 10 000 times
 - In real case with a large input file the comparison is 2h (using well implemented SAGA module) to 1 min.

Solution:

WASP editor (fix errors) -> SAGA (triangulation) -> Windmesh (Shepard)



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2. Targets



A perfect mesh

2. Targets

- Structured mesh whenever possible
- Mesh “following” wind flow
- Neighboring cell size difference should not be larger than 20%
- Fully automatic
- Covering large domain, with fine resolution and with a mesh of few elements.

2. Targets

- Simple geometry
 - A box with one “wavy side”.
 - No objects in the middle
- Inlet profile is important, and has to be maintained
 - Vertical resolution can not change
- Very fine horizontal resolution close to the ground
 - At least 10 layers below a hub

Very flat cells following terrain!

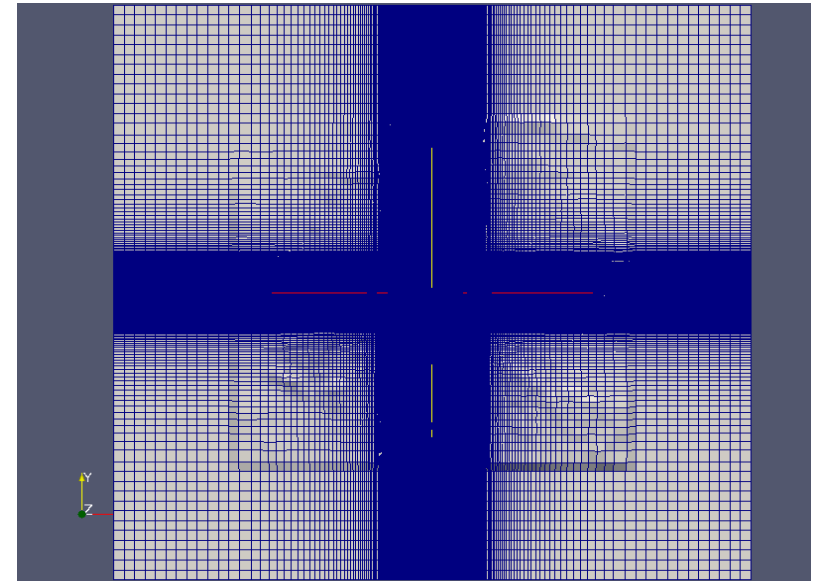
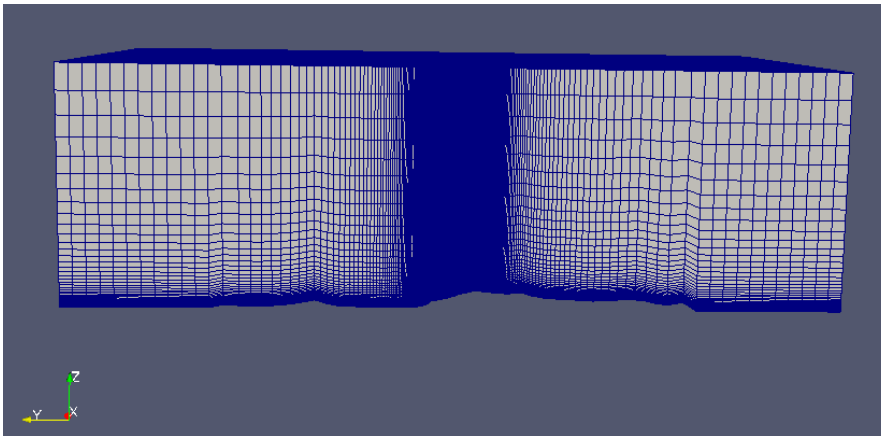
3. Mesh construction



Base mesh

3. Mesh construction

1. Create 2D mesh
2. Interpolate Z and Z0

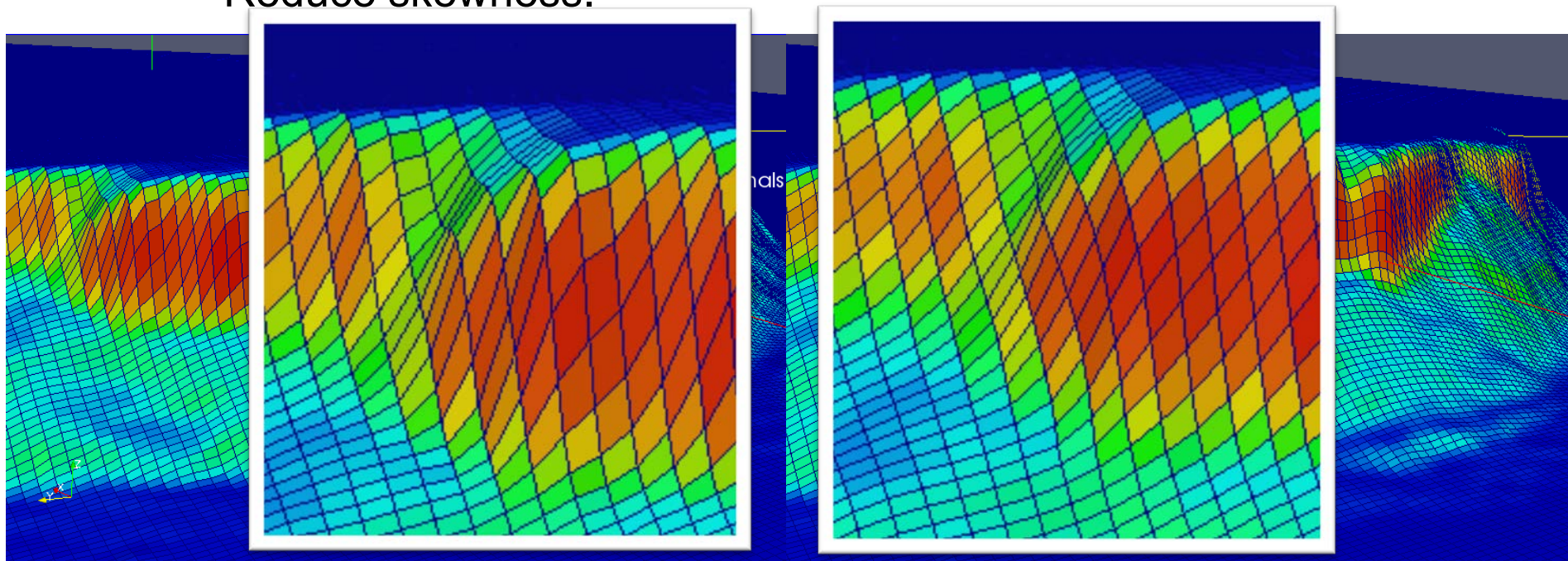


3. Create 3D mesh

Improving 2D mesh

3. Mesh construction

- Rotating the mesh to face the wind flow
- Make the ground faces “more square”.
 - Reduces the size difference of neighbouring cells
 - Reduce skewness.



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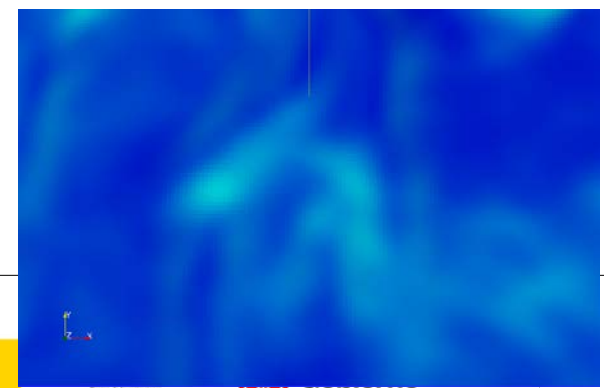
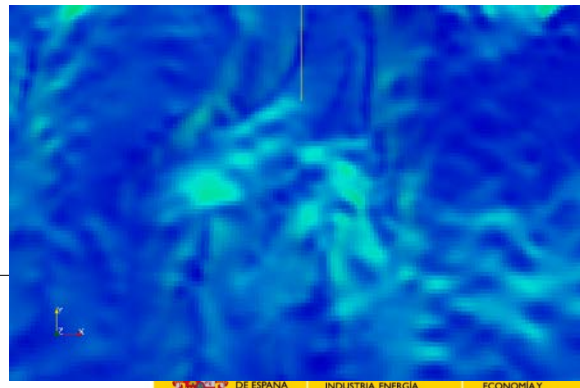
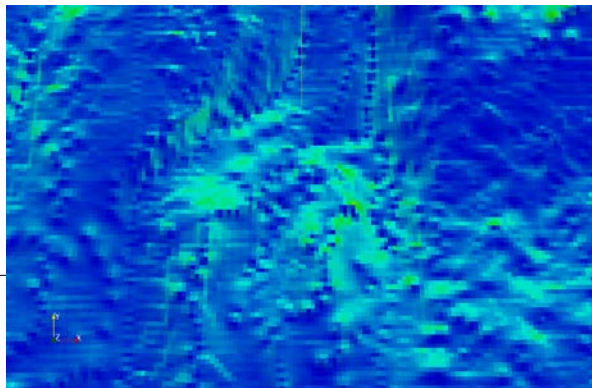
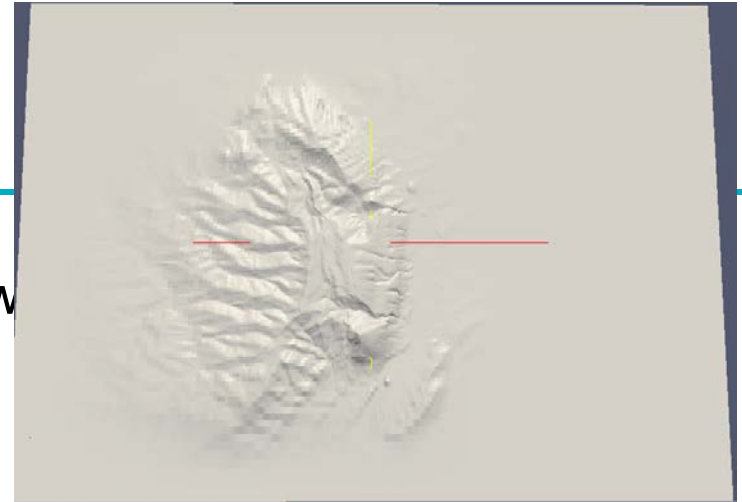


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Improving 2D mesh

3. Mesh construction

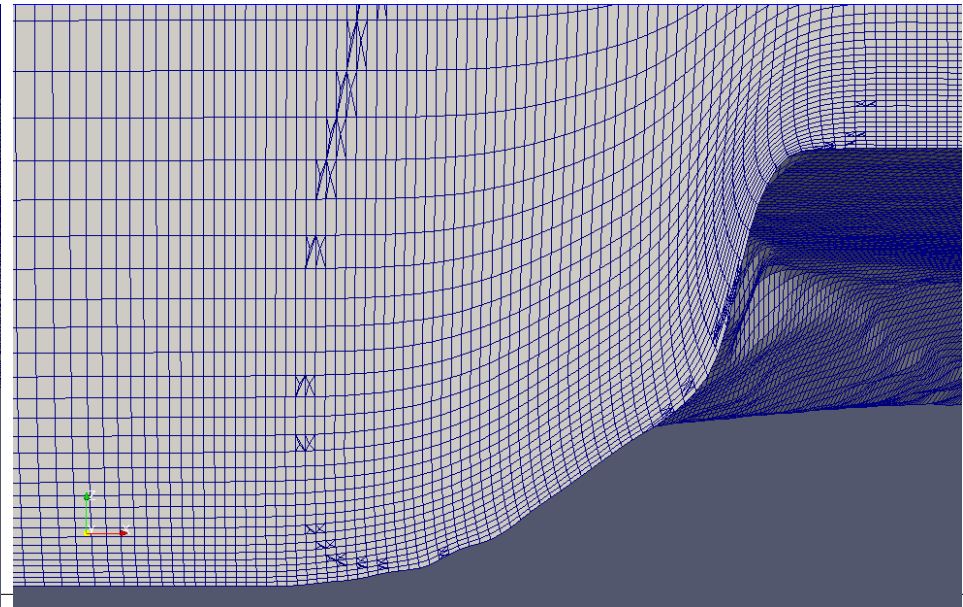
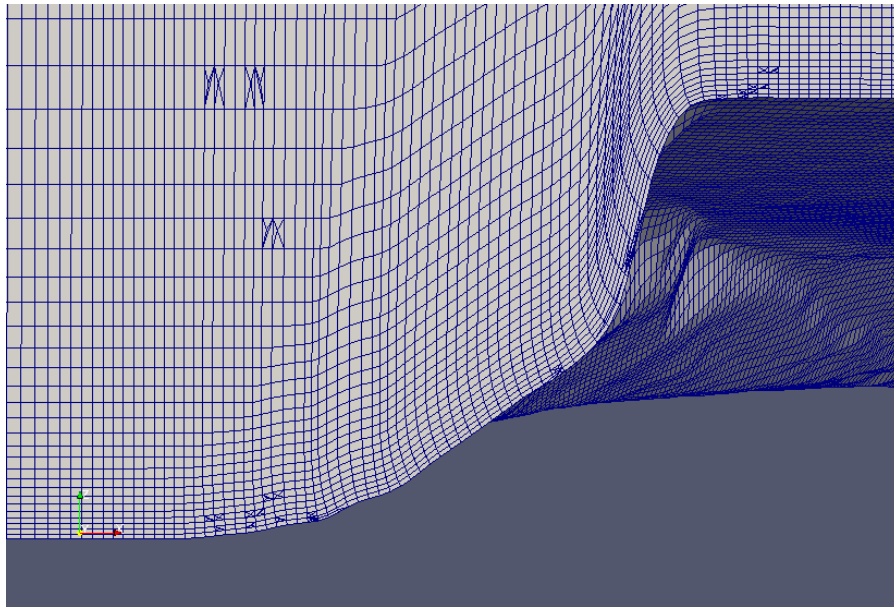
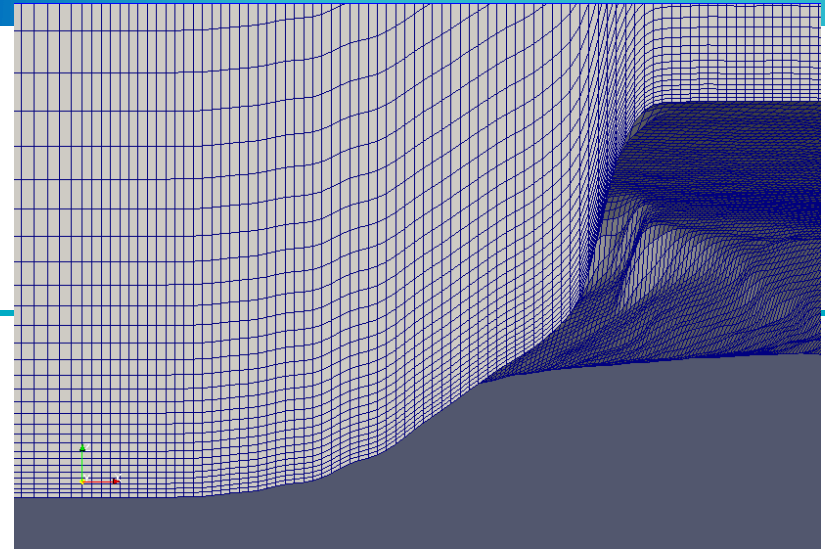
- Rotating the mesh to face the wind flow
- Make the ground faces “more square”.
- Smoothing
 - Best results when two approaches are used
 - Buffer and transition zone: cells are very large but flat. In those regions strong smoothing of the mesh is necessary to avoid convergence problems.
 - Farm zone: cells are smaller, real terrain is priority, depending on the input file noise can be present.



Improving 3D mesh

3. Mesh construction

- Cells should follow terrain.
- Cells above hills are flattened
- Elliptic smoothing
 - Improves mesh statistic but can not be used as a black box



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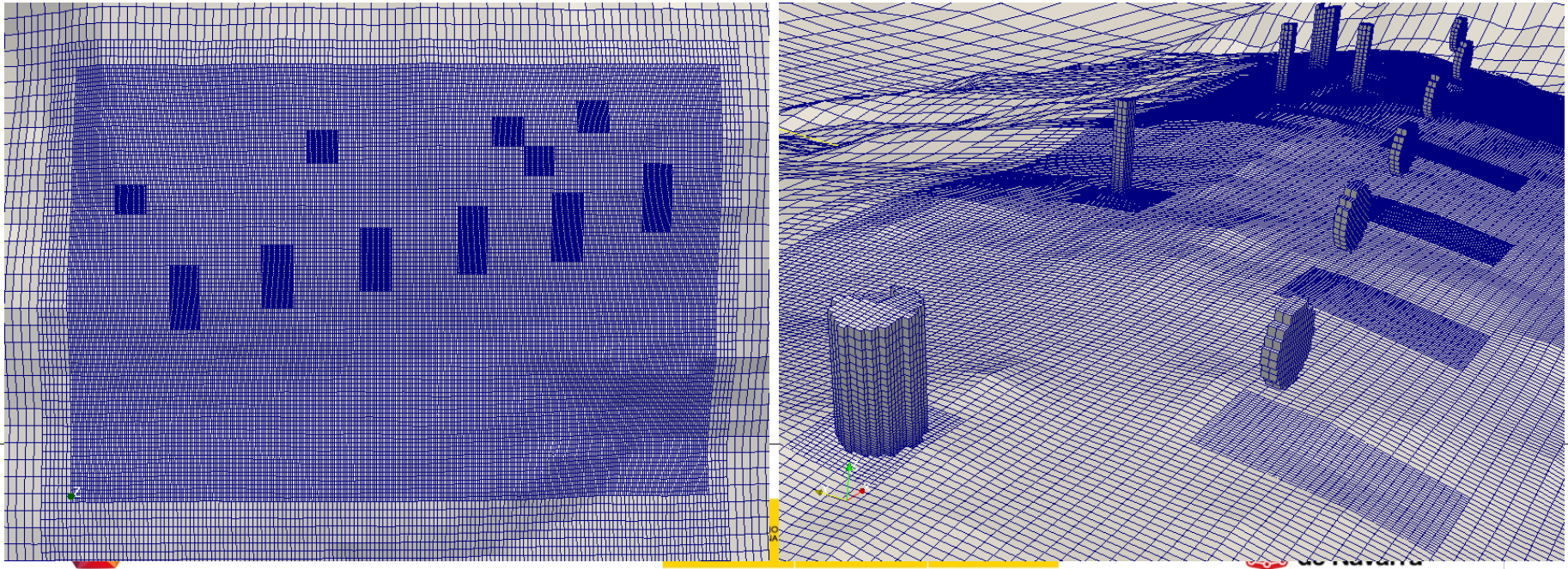


4. OpenFOAM tools

Improving 3D mesh

4. OpenFOAM tools

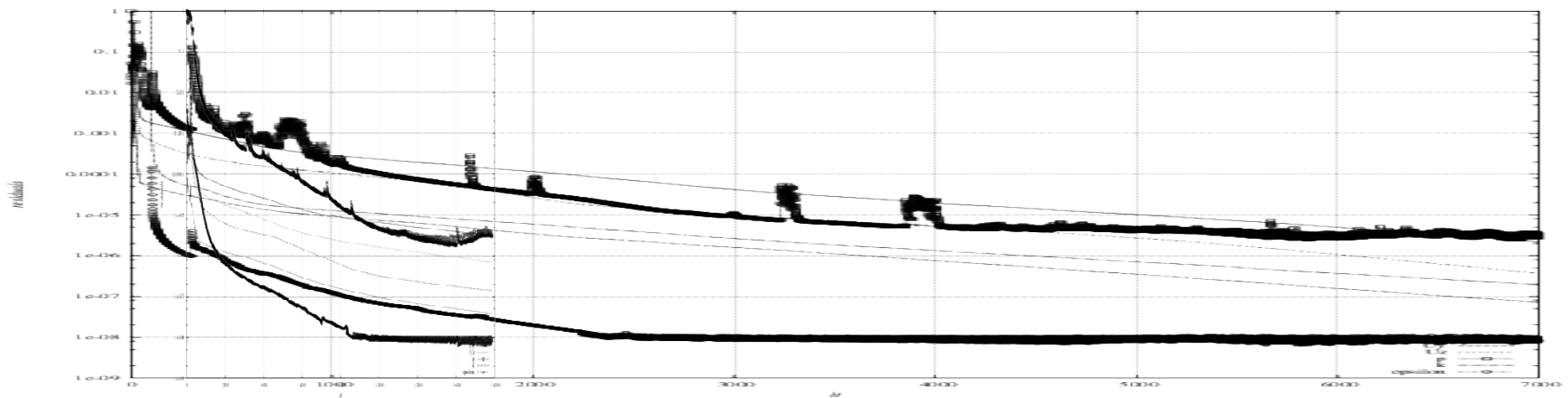
- topoSet
- refineMesh
 - Modelling wakes
- mapFields
 - Fine mesh can be initialised with results from a rougher one



Improving 3D mesh

4. OpenFOAM tools

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Conclusions

- .map files are prone to errors, difficult to process and inaccurate. If possible, they should be avoided.
- Delaunay triangulation should be used for problematic input, in other cases Shepard is more efficient.
- OpenFOAM tools allows adapting the mesh for wake models and give promising results in initialisation with rough mesh results.
- While the new meshing software allows models convergence systematic test are necessary in order to
 - improve results
 - see how can the mesh size be reduced



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